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ex 10.

Problem.

For any cut $(S, V \setminus S)$ in a connect weighted graph G, the minium weight edge.

base case: A tree on 2 vertex has 1 edge, which must cross every cut separating the two vertice. The edge is trivially the minium crossing edge, so it belongs to the MST.

Inductive hypothesis: Assume the cut property holds for all trees with fewer than k edges.

Inductive step: Let T be any MST of G with k edges, and let $e = (u, v)$ be the minium-weight edge crossing cut $(S, V \setminus S)$

Suppose for contradiction that $e \notin T$. Since T is a spanning tree, there exist a unique path P in T from u to v. Since $u \in S$ and $v \in V \setminus S$, this path must cross the cut so it constain at least one edge $e' = (x, y)$ crossing the cut.

Now construct $T' = T - \{e'\} + \{e\}$. Since e is the minium

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weight crossing edge:

$w(e) \leq w(e')$

So $w(T') \leq w(T)$. But T is an MST, so $w(T') \geq w(T)$ which means $w(T') = w(T)$ - T' is also an MST. This means we can always swap e' for e and obtain another MST containing e. Since T was an arbitrary MST, e belongs to every MST.

Problem 2.

(a) Hierarchical feedforward cascades.

Algorithm: Topological Sort (DFS-based)

justification: Information flows forward with no loops - this is exactly a DAG. Topological sort reveals the strict ordering of brain regions in the cascade.

time complexity: $O(V+E)$

(b) Recurrent feedback circuits

Algorithm: strongly connect components.

justification: recurrent circuits are closed loop where every region can reach every other - this is the definition of a strong component. SCC decomposition identifies exactly these

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self-sustaining feed back loops.
time complexity $O(V+E)$

c, ~~fath~~ fastest reactive pathway
algorithm: Dijkstra's algorithm.
justification: we want the minium-delay path from a stimulus region to an action region. ~~so~~ Since all weights are postive, Dijkstra correctly find the shortest path.
Time complexity: $O((V+E)\log V)$ with a priority queue.

Fundamental limitation.
static graph algorithm cannot model the brain's dynamic nature - connection strengths change ~~cannot~~ continuously with neural firing and plasticity, which a fixed graph ~~cannot~~ cannot capture.